

Rounak Sonavane

AI ENGINEER | AGENTIC AI | COMPUTER VISION

+91-7721008000 | rounaks7721@gmail.com

linkedin: rounak-sonavane | github: Rounak7721

PROFILE SUMMARY

AI/ML graduate specializing in Agentic AI systems, multimodal applications, and local-first AI engineering. Self-driven builder with hands-on experience developing planning agents, tool-orchestration systems, RAG pipelines, and intelligent automation workflows. Seeking AI Engineering, Agentic AI, and Applied AI roles.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, SQL

Agentic AI / LLM: LangChain, LangGraph, OpenAI API, Groq API, Ollama, RAG, n8n

AI/ML & Computer Vision: PyTorch, TensorFlow, OpenCV, MediaPipe, YOLO, InsightFace

Backend & Databases: Flask, REST APIs, PostgreSQL, MongoDB, DuckDB, FAISS, ChromaDB

Robotics & Tools: ROS2, MoveIt, Git, Linux, Docker

WORK EXPERIENCE

AI Intern

Jan 2025 – Apr 2025

Ramkrishna Foundry Pvt. Ltd.

Tech Stack: JavaScript, Express.js, LLM APIs, Google Sheets

- **Designed and developed** a company website with an integrated LLM-powered chatbot for visitor support, using Vite, JavaScript, and Express.
- **Built automated contact-form and data-capture workflows** with Google Sheets integration and web analytics tracking.
- **Supported exploratory AI integration initiatives**, evaluating practical LLM use cases for business operations.

PROJECTS

MindDesk — Multimodal Agentic AI Platform

[Rounak7721/MindDesk](#)

Python, Flask, React, Ollama, FAISS, ChromaDB, PostgreSQL, MongoDB, YOLOv11, InsightFace

Local-First, Fully Offline

- **Architected a custom, fully local Plan-and-Execute agent engine** running entirely on private hardware (no cloud dependency), unifying 8 capability domains behind one natural-language interface and orchestrating 17 modular tools via a decoupled tool registry.
- **Built an autonomous self-correction loop** with a 3-attempt retry mechanism that uses the reasoning model to repair failed tool-call arguments without human intervention.
- **Implemented fully local RAG pipelines** (FAISS + ChromaDB) for PDF, DOCX, XLSX, and CSV ingestion and semantic retrieval, running entirely offline using locally hosted LLMs.
- **Engineered a polyglot persistence layer** spanning 5 isolated PostgreSQL schemas, MongoDB, and vector stores, plus a real-time YOLOv11 + InsightFace pipeline for surveillance and facial recognition, and natural-language Arduino/IoT device control.

Agentic Lead Qualification System

[Rounak7721/ai-lead-qualification-system](#)

n8n, LLaMA 3.1 (Groq/Ollama), Airtable, Google Sheets, Slack

- **Designed an end-to-end automated agent workflow** in n8n that ingests leads, scores intent on a 1–100 scale via LLM reasoning, and routes them to Slack/CRM by priority tier with zero manual triage.
- **Built idempotent, stateful pipelines** across real-time intake and scheduled follow-up paths, integrating Airtable CRM and an append-only Google Sheets audit log, with hybrid cloud (Groq) and local (Ollama) LLM deployment support.
- **Implemented robust LLM-output parsing** with fallback logic to handle malformed model responses reliably in production.

IRIS — Intelligent Robotic Perception & Manipulation System

[Rounak7721/iris](#)

Python, C++, ROS2, MoveIt, OpenCV, YOLO

Ongoing

- **Building a 6-DOF robotic arm system** integrating ROS2 + MoveIt motion planning with a dual-camera perception pipeline for object tracking and world-state modeling.
- **Designing an agentic planning layer** to translate high-level natural-language goals into autonomous manipulation task sequences, combining YOLO-based detection with classical vision techniques.

Virtual Mouse — Gesture-Controlled HCI System

[Rounak7721/Virtual-Mouse](#)

Python, OpenCV, MediaPipe

- **Built a real-time, touchless gesture-control system** tracking 21 MediaPipe hand landmarks to replace mouse input with cursor movement, clicks, scrolling, and drag-and-drop gestures.

EDUCATION

Bachelor of Technology in Artificial Intelligence & Machine Learning

2021 – 2025

Sanjay Ghodawat University